

WORK FORCE DEVELOPMENT AUTHORITY (WDA)

GITISI SECONDARY SCHOOL (AFPRO / GSS)

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THREADING OF PIPE

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CHAP 1. GETTING STARTED WITH PLUMBING

I.INTRODUCTION TO PLUMBING

Plumbing is one of the most important services to be found in any modern society.

For people in society to lead a healthy life, one of the most basic requirements is to live in clean and sanitary conditions. This means having access to a piped supply of clean water, and accommodation that includes appliances or fittings for washing, cleaning and the collection and disposal of waste matter, therefore Plumbers who install and maintain such systems, are seen as important figures in society.

Modern plumbing can seem and often is complicated and the modern plumber needs to be well trained or skilled to cope with all the different things that he or she has to do.

It was probably the Romans however who had the most influence on the development of plumbing. The word ‘plumber’ itself is taken from the Latin word for lead, plumbum , Plumbers were originally seen as workers of lead.

Plumber must also be able to read and interpret details contained within a number of information sources including drawings, specifications and manufacturer’s catalogues.

Generally, we think that plumbing meets people’s basic requirements by:

- Keeping them healthy and clean by providing cold water for drinking, hot water for washing and sanitation systems for the removal of waste products
- Keeping them warm with the help of hot water heating systems
- Meeting the above requirements together to ensure maximum comfort and convenience for them by providing heating, hot and cold water systems and sanitation 365 days a year and 24 h a day.

Explore the plumber’s role a little further and you will see that most plumbers usually carry out the:

- Installation

- Service
- Maintenance of a wide range of domestic systems such as:
 - ✓ Cold water, including underground services to a dwelling
 - ✓ Hot water
 - ✓ Heating systems fuelled by gas, oil or solid fuel
 - ✓ Sanitation (or above ground drainage) including the installation of baths, hand wash basins, water closets (WCs) and sinks
 - ✓ Rainwater systems, gutters and fall pipes
 - ✓ Associated electrical systems
 - ✓ Sheet lead weathering (at Level 2, this includes things, such as chimney weathering, and soil vent pipe weathering).

In fact, you will see that the above list matches the content of the Technical Certificate. In addition to the systems, the plumber will also have to work on the appliances and components contained within them.

Here are a few examples:

- Storage vessels, cylinders, cisterns
- Sanitary appliances, sinks, baths, WCs, washbasins
- Domestic appliances including washing machines and dishwashers
- Heat exchangers, boilers, radiators
- Pumps, accelerators and motorized/isolating valves
- Gas appliances (natural or Liquefied Petroleum Gas (LPG), boilers, water heaters, cookers and fires)
- Pipe materials, fittings, fixings, controls that constitute the above systems.

Plumbers also work on:

- Cabling and electrical components, only if qualified to do so
- Sheet weathering, aprons, back gutters, step flashings, soakers, lead slates.

So, we are now beginning to build up a picture of the extent of a plumber's job. What skills and knowledge do you think a plumber needs in order to be able to carry out the job competently?

Here are our thoughts on what a competent plumber should be able to do:

- Follow health and safety legislation and guidance at all times
- Thoroughly plan the job including making sure that all the tools, materials and equipment are present on the job
- Agree a schedule of work with the customer or client
- Provide a cost estimate for the job
- Prepare the work location, making sure that there is adequate access
- Protect the customer's property
- Mark out, measure and work out the installation requirements
- Fabricate, position and fix system components
- Pre-commission (including testing), commission and de-commission systems
- Service and maintain system components
- Work effectively with customers, workmates and other site visitors
- Work in an environmentally friendly manner
- Promote the products and services of the plumbing business.

We also think that the plumber should have a working knowledge of:

- Regulations
- Codes of practice
- Principles of plumbing systems including basic design
- Where to find manufacturers' technical data
- Health and safety legislation and guidance
- Commercially agreed standards.

II .SAFETY AND PRECAUTION ON WORKPLACE

Safety can be defined as the protection from a hazard.

It is widely known that the construction industry as a whole, of which plumbing is a part, accounts for many of the accident statistics recorded.

Protective clothing

The safety equipment is defined as all equipment that are designed to be worn or held to protect against risks to health and /or safety. Personal protective equipment includes most types of protective clothing and equipment, such as eye, hand, foot and head protection

Eye protection

Your eyes are two of your most precious possessions. You need them for learning and for earning, because without your eyesight you probably cannot perform your present job. Your eyes are extremely vulnerable to injury at work.

There are many types of eye protection equipment available:

- Safety glasses
- Safety goggles
- Full face masks/shields (is used even to protect nose and mouth)

Foot protection

The basic universal form of foot protection is the safety shoe which will usually include some kind of metallic toe protection , rubber soles and boots.

Hand protection

Your hands, which you use for almost everything including working, playing, driving, eating....Unfortunately, hands are often injured. Almost one in four work-related injuries happens to the hands and fingers. One of the most common problems other than cutting, crushing or puncture wounds to the hand is dermatitis. Dermatitis is an inflammation of the skin normally caused by contact with irritant substances.

The irritation of the skin may be indicated in several ways, sores, blisters, redness or dry cracked skin, which easily become infected. To protect your hands from irritating substances, you need to keep them clean by regular washing using approved cleaners. Make good use of barrier creams where provided , and wear appropriate personal protection(usually a strong pair of gloves).

Head protection

This is another important form of personal protective equipment and is especially important to plumbers who need to spend time on rooftops and up ladders. A hammer accidentally dropped

from a roof location could cause serious injury or death if it landed on someone's unprotected head.

Hard hats constructed from high-density polystyrene, will provide some protection from falling objects and accidental head knocks.

Head protection is extremely important because it guards your most vital organ (your brain). As we have previously said, a head injury can physically disable a person for life and can even result in death. Head protection can help prevent such injuries. Approximately 80% of industrial head injuries are a result of people not **wearing hard hat** protection. You need to wear your safety helmet whenever it is required and should ensure that it is worn correctly

1.3. Plumbing tools

Introduction

You will use a wide variety of tools and equipment as a plumber, all of which can be potentially dangerous if misused or neglected. You should always use the right tool for the job; never just make do with whatever tool you may have at hand.

Example: screwdrivers are not intended to be used as an alternative to wood chisels or bolsters for lifting floorboards.

As a plumber, you will be required to Measure; Mark out; Cut ;Fabricate ;Make joint; and ;Fix a range of materials.

COMMON CATEGORIES OF TOOLS USED IN PLUMBING

- hand tools for Cutting
- Marking tools
- Measuring tools
- Fixing(holding) tools
- Hammering tools
- power (special)tools

In most of the cases, this will involve the use of tools to enable you to do this.

Once you have got your tool kit, keep the tools clean and well maintained, this should ensure a long life and keep down the cost of having to regularly buy replacements.

1. MEASURING TOOLS

The standard tools include:

- Tape Measure

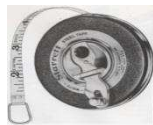
- Spirit Level

- Folding Steel Rule
- Try square
- Combination square
- Vernier caliper
- Framing square

(i)



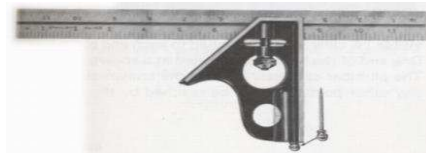
Tape measure



100 foot steel tape used for measuring long runs of pipe



Try square

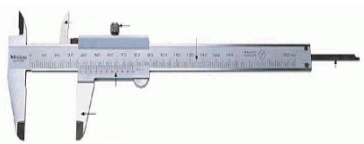


Combination square



Framing square; is used for measuring, squaring, and marking cuts to be made on walls and partitions

Vernier caliper



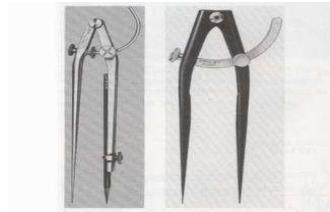


spirit level

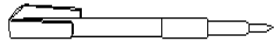
2. MARKINGOUT

Marking tools includes:

- Pencil
- Scriber
- Center punch
- Compass and divider



Compass and divider



Scriber



Center punch

And do not forget, you will need a pencil for marking out!

3. *Cutting and preparation*

This will require:

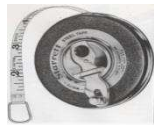
- Hacksaw Frame Straight Tin Snips
- Compact Pipe Cutter
- Trimming Knife
- File

- Wood Chisels
- Floorboard Saw
- Abrasive Cleaning Pads
- Cold Chisel with Guard
- Floorboard/Brick
- Bolster with Guard
- Pad saw
- Junior Hacksaw.

CUTTING TOOLS:



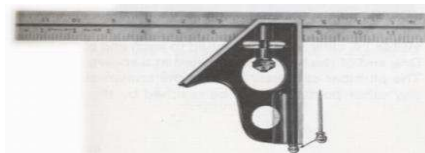
Tape measure



100 foot steel tape used for measuring long runs of pipe



Try square

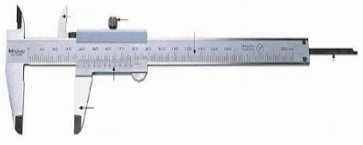


Combination square

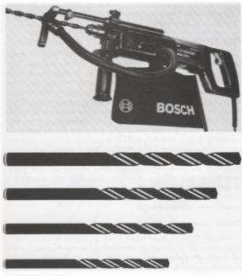


Framing square; is used for measuring, squaring, and marking cuts to be made on walls and partitions

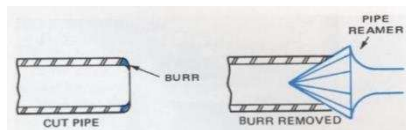
Vernier caliper



Pipe cutters



Rotary hammer drill with drill bits



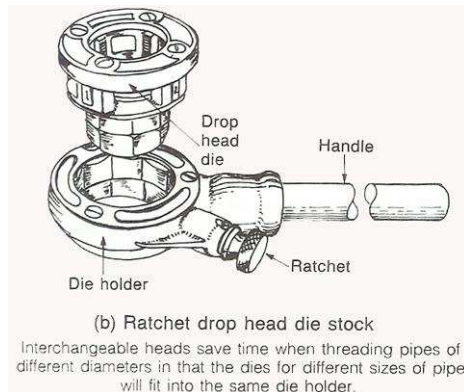
Reamers



Dies



(a) A three-way pipe die and stock permits three diameter of pipe to be threaded with a single tool.



4. Fabrication

Bending and threading equipment is also available for use with low carbon steel (LCS) pipe.

Fabrication tools include:

- Claw Hammers
- Floorboard/Nail Bar
- Club/Lump Hammer
- Copper Pipe Hand Bender
- Internal Bending
- spring
- External Bending

- Spring.

hammering tools



claw carpenter's hammer



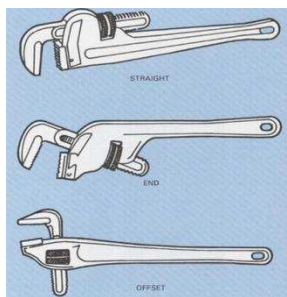
ball peen hammer

5. FIXING (HOLDING) TOOLS

- (a) Adjustable Basin Wrench
- (b) Adjustable Wrench
- (c) Basin Wrench
- (d) Adjustable spanner
- (e) One Hand Speed Wrench
- (f) 250mm Waterpump Pliers
- (g) Blow Lamp-Propane Torch
- (h) Combination Pliers
- (i) Curved Jaw Locking Plier

As you build up your tool kit over time, you may consider investing in a set of spanners.

- (j) Spanners



pipe wrench



Chain wrench



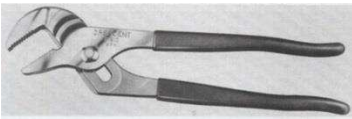
A monkey wrench, is like a pipe wrench with smooth jaws



Open end spanner



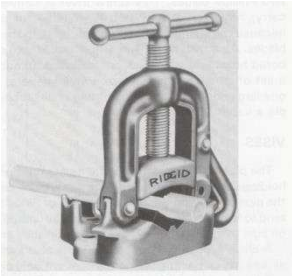
Adjustable wrench, handles nuts of all sizes up to its capacity



Tongue-and-groove pliers



. Locking pliers exert extraordinary clamping pressure



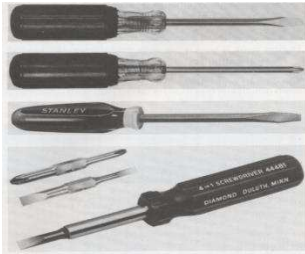
Pipe vise holds pipe for other operations being performed



Chain-type vise serves same purpose as regular pipe vise

Fixing and making good

- 8 piece screwdriver set
- Insulated screwdriver
- Flat piece wood bits set
- 150 mm pointing trowel
- Masonry drill bits
- High speed steel drill bits.



Straight blade and Phillips screwdrivers in several sizes

Other tools

- Allen keys
- Immersion heater key
- Sink plunger
- Stop valve key
- Tool box.

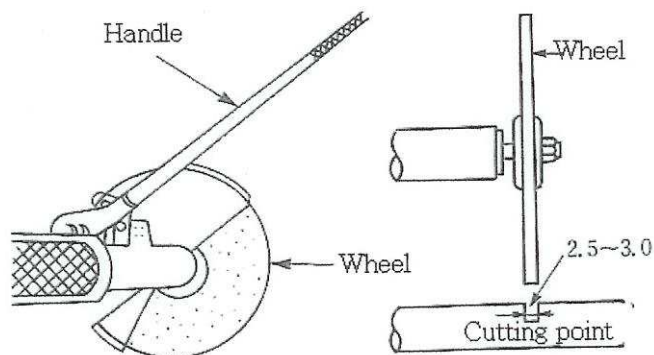
6. Specialist tools

Plumbers also use tools and equipment of a more specialist nature and these should be used in accordance with manufacturers' instructions and by personnel who have been properly instructed in their use (Figure 3.3).

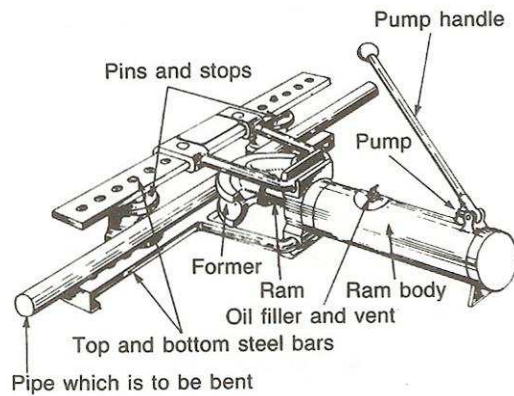


Figure 3.3 Pipe freezing kit

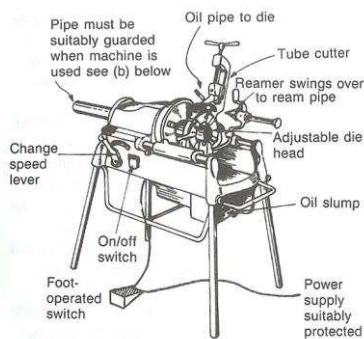
SPECIAL TOOLS



High speeds wheel cutting machine

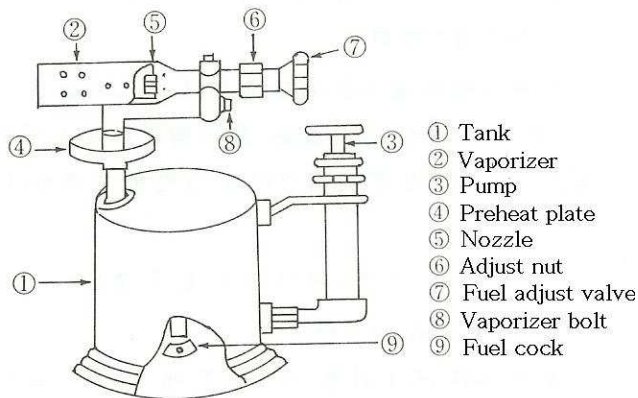


Hydraulic press bender/pipe bending machine



(a) Portable power threading machine

Power threading machine



Torch lamp

In addition to the tools and equipment, you will need materials such as:

- Solder wire
- Fluxes approved for plumbing work (including those suitable for wholesome water installations)

- Jointing tape approved for plumbing work
- Jointing compound approved for plumbing work (including those suitable for wholesome water installations)
- A range of screws and nails.

Power tools

Cordless power drills and screwdrivers

The cordless drills/screwdrivers are available in a range of voltages; 12, 14, 14.4 and 24 V being a few examples. Most drills are combined so they can be used as a drill and a screwdriver.

They are powered by batteries (usually supplied with two) and a charger, so you can have one working and one on charge.

Power drills 110V

There is a wide range of makes and models of power drills; this section is designed to show a cross section of what is available.

Typical power ratings for 110 V drills range from 620 to 1400W. Most drills are of variable speed and some have a reversible action. Drills vary in power depending on the size of motor; this in turn has a bearing on what the drill can do.

The one shown in Figure 3.5 has a ‘hammer action’ which when engaged makes drilling through masonry easier. Not only does the drill rotate but it also moves fractionally backwards and forwards at high speed, giving the effect of hammering the drill into the building fabric.

Power saws

Power saws such as circular saws are used by plumbers for taking up floorboards or sheets in order to install pipework under floors

Power tools in general

Here are a few points to remember about power tools:

- All electric tools should be double insulated
- Always use 110 V supply
- Check that electrical cables are not damaged or worn out

- Check that plugs are not damaged
- Check for test labels to show the equipment is safe to use
- Remember to wear safety goggles when using drills and saws. These will protect your eyes from dust and any splinters of material that might fly off whilst working

MATERIALS USED IN PLUMBING

4.1. PROPERTIES OF MATERIALS

What are a material's properties?

In general terms, you may say this could be how strong it is, how well it conducts heat or electricity, or how flexible it is.

Scientifically, materials are classified according to a variety of properties and characteristics. The properties can be measured as the materials react to a variety of influences, which include:

- Mechanical properties, such as hardness, strength, elasticity, toughness, stiffness, ductility, malleability
- Thermal properties like conductivity (how well or poorly a material will conduct heat)
- Electrical properties like conductivity (how well or poorly a material will conduct electricity)
- Chemical properties like reactivity and solubility
- Optical properties like transparency, reflectivity, and refractivity
- Magnetic properties.

Hardness

There are many different aspects of materials which could be considered as a measure of hardness. Hardness can mean resistance to permanent or plastic deformation by scratching, indentation, bending, breaking, abrasion or fracture. This is a very important factor in materials which have to resist wear or abrasion – a sink tap for example – and frequently needs to be considered along with the strength of materials.

Strength

The strength of a material is the extent to which it can withstand an applied force or load (stress) without breaking. Load is expressed in terms of force per unit area, and is measured in newtons per square metre (N/m²). This can be in the form of:

- Compression force, as applied to the piers of a bridge, or a roof support

- Tensile or stretching force, as applied to a guitar string, tow rope or crane cable
- Shear force, as applied by a shearing machine or scissors, or when materials are torn (see below). Materials are therefore described as having compressive, tensile or shear strength.

Hardness is measured on a scale, Mohr's scale of 1 (talc) to 10 (diamond)

Elasticity

Almost all materials will stretch to some extent when a tensile force is applied to them. This increase in length on loading, compared to the original length of the material, is known as strain.

As increased loading continues, a point is reached when the material will no longer return to its original shape and size on removal of the load, and permanent deformation has occurred. The material is said to have exceeded its elastic limit or yield stress, beyond which the material is suffering plastic deformation – it is being stretched irreversibly.

Here are how some common materials 'shape up'

- Mild steel has little elasticity, but has a high yield stress and is fairly ductile, i.e. has a large range over which it can sustain plastic deformation. It also has a high tensile strength.
- Cast iron is brittle – it has poor elasticity and has no ability to sustain plastic deformation, although its tensile strength is higher than that of concrete.
- Copper has little elasticity, but is ductile. It has an ultimate tensile strength less than half that of mild steel.
- Concrete has little elasticity, and the lowest tensile strength of the four, but is extremely strong in compression.

Some other important characteristics which must be considered when considering material used in the plumbing trade are:

- ***Plasticity*** The exact opposite of elasticity: a material which does not return to its original shape when deformed
- ***Ductility***: Is the ability of a material to withstand distortion without fracture, an example is a metal such as copper that can be drawn out into a fine wire
- ***Durability*** :The material's ability to resist wear and tear
- ***Fusibility***: The melting point of a material, i.e. when a solid changes to a liquid
- ***Malleability*** :The ability of a metal to be worked without fracture; sheet lead is a very malleable metal
- ***Temper***: The degree of hardness in a metal
- ***Tenacity***: A material's ability to resist being pulled apart
- ***Thermal*** :The amount a material expands when expansion heated.

By now you should have an idea of the basic properties of material.

Next we are going to look at plumbing pipework materials. There is no perfect pipework material that is suitable for all applications; different materials perform better in relation to different factors and conditions which can affect pipework such as:

- Pressure
- Properties of the water
- Cost
- Bending and jointing method
- Corrosion resistance
- Expansion
- Appearance.

There are two basic types of pipework material: metal and plastic.

Metals commonly used in the plumbing trade

Metals used in the plumbing industry include steel, iron, copper, brass, lead, tin, zinc and aluminium.

Copper

Copper is supplied in lengths and coils and in a range of diameters. Copper is a malleable and ductile material which you will use frequently throughout your plumbing career. There are several types of copper tubes manufactured for use in the plumbing trade:

- Soft annealed coils : is a copper tube used for microbore pipework (usually found on central heating systems) and can be bent by hand or bending machine
- Half hard straight length: is the copper tube most commonly used above ground for most plumbing and heating installations. It is fairly rigid and usually needs to be bent using a bending machine or spring.
- A more rigid thin-walled copper tube: is also available. The tube is mainly used for export purposes and cannot be bent.
- Tubes complying with the above specifications are not intended to be buried underground.
- Thick-walled soft annealed coil: is most commonly used underground for the conveyance of water.

The outside diameter of the tube is the same for each type, but there are differences in the internal bore size due to the variations in wall thickness.

Lead

The term plumber is derived from the chemical symbol Pb, and the Latin phrase plumbum, which when translated, means ‘worker of lead’. Traditionally, lead was commonly used for water

supply, sanitary and rainwater pipework, but it has now been superseded by the use of materials, such as plastics and copper.

These days, its main use in the plumbing industry is for sheet lead weathering's as its use for new water supply pipework is prohibited, although you may come into contact with lead pipework if renewing an old service pipe.

Lead is a very heavy, valuable metal which requires careful handling. It is ideal for sheet roof work as it is extremely malleable; with the 'bossing' of an internal corner.

Cast iron

Cast iron is an alloy of iron and approximately 3% carbon. It has been used in the plumbing trade for many years for above and below ground drainage pipework. *Cast iron* is very heavy but quite brittle, and can withstand many years of general wear and tear. You will probably come into contact with it on older properties – on newer buildings it has been superseded by *plastic-PVC*.

Copper is also used in the manufacture of pipe work fittings .As with *lead*, cast iron is sometimes used on historic buildings and unlike lead, *cast iron* is occasionally used on new build public and commercial buildings, where its strength and rigidity are an *advantage*. Cast iron was used in the manufacture of baths

Alloys

Alloys can be produced either by mixing different metals or by mixing metals with non-metallic elements, such as carbon. ***Steel*** is one of the most common alloys in the world and many plumbing materials and appliances are made of different types of steel.

Other alloys include:

- Brass (alloy of copper and zinc) which is used for various types of plumbing fittings . Taps are chrome plated for decorative effect.
- The image shows an example of brass used in the manufacture of pump isolation valves.
- Bronze (copper, and tin with or not zinc), again, used mainly for fittings
- Solder (lead and tin), which is used for soldering capillary fittings, although solder containing lead is no longer permitted for use on water supply pipework, which is a requirement of the Water Regulations.

Low carbon steel

Low carbon steel (LCS) or *mild steel* is an alloy made from iron and carbon. It is commonly used in the plumbing and heating industry in larger premises, such as factories and commercial buildings.

In the domestic market, radiators are manufactured from steel, and LCS pipework and fittings are used in some small residential buildings for central heating systems.

LCS pipe is manufactured to BS 1387:1985 and comes in three grades of weight: light, medium and heavy. As with *copper* tube, the internal bore and wall thickness varies.

- Light LCS tube thin walls larger bore
- Medium LCS tube medium walls medium bore
- Heavy LCS tube thick walls smaller bore

CHAPTER 2: WATER SUPPLY AND VARIOUS TYPES OF PIPES

1 INTRODUCTION

Water supply is the process of general requirement for supply of water from public water supply system to individual building and subsequent distribution of water to various parts of the building. The water from public water supply system to individual buildings is supplied through pipes. A large proportion of capital is invested on pipes while designing water supply distribution system.

A **pipe** is a tubular section or hollow [cylinder](#), usually but not necessarily of [circular cross-section](#), used mainly to convey substances which can flow — liquids and gases (fluids), slurries, powders, masses of small solids. It can also be used for structural applications; hollow pipe is far stiffer per unit weight than solid members.

In common usage the words *pipe* and [tube](#) are usually interchangeable, but in industry and engineering, the terms are uniquely defined. Depending on the applicable standard to which it is manufactured, pipe is generally specified by a nominal diameter with a constant outside diameter (OD) and a schedule that defines the thickness. Tube is most often specified by the OD and wall thickness, but may be specified by any two of OD, inside diameter (ID), and wall thickness. Pipe is generally manufactured to one of several international and national industrial standards.^[1]

While similar standards exist for specific industry application tubing, tube is often made to custom sizes and a broader range of diameters and tolerances. Many industrial and government standards exist for the production of pipe and tubing. The term "tube" is also commonly applied to non-cylindrical sections, i.e., square or rectangular tubing. In general, "pipe" is the more common term in most of the world, whereas "tube" is more widely used in the United States.

Both "pipe" and "tube" imply a level of rigidity and permanence, whereas a [*hose*](#) (or hosepipe) is usually portable and flexible. Pipe assemblies are almost always constructed with the use of [*fittings*](#) such as elbows, tees, and so on, while tube may be formed or bent into custom configurations. For materials that are inflexible, cannot be formed or where construction is governed by codes or standards, tube assemblies are also constructed with the use of tube fittings.

The following factors should be considered in selection of pipes.

- Strength of pipe
- Water carrying capacity
- Life and durability of pipe
- Expenditure on transportation
- Jointing process, maintenance and repairs.

2VARIOUS TYPES OF PIPE

The pipes are available in several types and sizes. They may be classified into three groups:

- Steel pipes
- Copper pipes and
- Plastic pipes.

3.1.1 STEEL PIPES

These pipes are very extensively used for water supply. These are suitable for long distance pipe lines of high pressure and provide satisfactory performance during service.

Steel pipes have excellent mechanical properties and are ideally for welding. During transportation, there is a minimum damage in pipes.

3.1.2 COPPER PIPES

Copper pipe and tube comes in a variety of types, with different wall thicknesses, ductility and intended used. The difference between copper pipe and copper tube is the the way the diameter

of the pipe is measured. Copper tube is measured by outside diameter (OD) whereas copper pipe is measured by inside diameter (ID). Depending on the plumbing job you are doing, local and national plumbing codes will dictate which type of copper pipe is acceptable.

Copper pipes come in four types - type K, type L, type M, and DWV. Type K has the thickest walls and is most frequently used for underground service lines in the supply system. Type L is used for interior hot and cold water supply lines. Type M is the thinnest of the types and is also used for interior hot and cold water supply lines. DWV pipes are used for the drain-waste system and for the vent system.

I. Type L copper pipe

Type L copper pipe and tube has a thicker wall than type M and DWV pipes making it the preferred choice for longevity. There are two kinds of type L; Hard, and soft temper. Type L will be marked with blue along the pipe or tubing.

Hard temper type L plumbing applications include:

- Above ground water distribution
- Above and below ground drainage and venting systems
- Building sewer

Soft temper type L plumbing applications include:

- Water service pipe
- Water distribution above and below ground

II. Type M copper pipe

Type M copper pipe and tubing is commonly used in residential plumbing because it has thin walls and can be produced and sold at a much lower cost. For water distribution longevity type M is not recommended. Type M copper is also better for heating applications because of the thin wall thickness. Type M is identified with RED markings along the pipe.

Hard temper type M plumbing applications include:

- Above ground water distribution
- Above ground drainage systems

Soft temper type M shall not be used in plumbing systems.

III. Type K copper pipe

Type K copper pipe and tube is the most robust of the four types because it has the largest wall thickness. Type K comes in hard and soft temper and will be identified by green markings. Type

K copper can be used for many other applications such as: Fuel, gasses, HVAC, fire protection systems and vacuum systems to name a few.

Hard temper type K plumbing applications include:

- Above ground water distribution
- Above and below ground drainage and venting systems
- Building sewer

Soft temper type K plumbing applications include:

- Water service pipe
- Water distribution above and below ground

IV. DWV copper pipe

DWV copper pipe is used for drainage waste and vent (DWV), above ground only and is identified by yellow markings.

3.1.3 PLASTIC PIPES

Plastic pipes are easy to work with, light-weight, and durable. Because of these reasons, plastic pipe has become the most popular pipe for do-it-yourselfers to use and there are five main types, each designed for specific applications.

PVC pipes are white and are approved for cold water use only. PVC - DWV pipes are also white, but are approved for use in drain-waste systems or vent systems. CPVC pipes are beige and are approved for use in hot or cold water service lines. Polyethylene tubes are black and approved for cold water use. And Polybutylene is gray or beige and is usually sold in coiled lengths. Polybutylene is the only flexible pipe approved for hot and cold service lines.

3 TYPES OF PLUMBING PIPES USED FOR RESIDENTIAL APPLICATIONS

There are many types of plumbing pipes used for residential applications. These include **PEX Tubing/Piping, Copper Pipes, PVC (Polyvinylchloride) and ABS Pipes**. Pipes are the heart of any plumbing system. They are the most important parts as they deliver the water; without them, nothing works! If you are unsure about which pipes are best for your home, then [find a professional plumber](#) who can surely give you some good advice.

- **PEX Tubing/Piping** - Pex Tubing/Piping is flexible plastic piping which has become the most popular type of piping used in U.S. homes today for some really great reasons. They are cost effective, easy to install, require little maintenance, less chances of leaking, resists buildup and are energy efficient. They can also be used for hot and cold water supply but cannot be used in outdoor applications due to their sensitivity to UV rays.
- **Copper Piping** - Copper pipes are second best next to PEX piping and can be used for both hot and cold water supply. They are corrosion resistant, durable and lightweight and

generally perform well. The down side is that installation is messy as the pipes must be soldered together which also requires numerous fittings.

- **PVC (Polyvinylchloride)** - These pipes are only for cold water and are used most often to carry water from the main supply to a home. PVC pipes must also be approved outdoor for drinking water.

ABS (acrylonitrile-butadiene-styrene) - is a black colored pipe and used for waste removal, venting, and drains primarily in bathrooms or mobile homes. It is a rigid plastic pipe that can be connected to metal pipe.

CHAPTER 3 FITTINGS FOR BOTH PIPING AND PLUMBING

1 INTRODUCTION

Fittings are used in pipe and plumbing systems to connect straight **pipe** or **tubing** sections, to adapt to different sizes or shapes, and for other purposes, such as regulating or measuring **fluid** flow. The term *plumbing* is generally used to describe conveyance of water, gas, or liquid waste in ordinary domestic or commercial environments, whereas *piping* is often used to describe high-performance (e.g. high pressure, high flow, high temperature, hazardous materials) conveyance of fluids in specialized applications. The term *tubing* is sometimes used for lighter-weight piping, especially types that are flexible enough to be supplied in coiled form.

Fittings (especially uncommon types) require money, time, materials, and tools to install, so they are a non-trivial part of **piping** and **plumbing** systems. **Valves** are technically fittings, but are usually discussed separately

2 Common fittings for both piping and plumbing

While there are hundreds of specialized fittings manufactured, some common types of fittings are used widely in piping and plumbing systems.

- **Elbow**



Short radius or regular 45° elbow (copper sweat)



Long radius or sweep 90° elbow (copper sweat)

An elbow is a pipe fitting installed between two lengths of pipe or tubing to allow a change of direction, usually a 90° or 45° [angle](#), though 22.5° elbows are also made. The ends may be machined for [butt welding](#), [threaded](#) (usually [female](#)), or [socketed](#), etc. When the two ends differ in size, the fitting is called a **reducing elbow** or **reducer elbow**.

Elbows are categorized based on various design features as below:

- Long Radius (LR) Elbows – radius is 1.5 times the pipe diameter
- Short Radius (SR) Elbows – radius is 1.0 times the pipe diameter
- 90 Degree Elbow – where change in direction required is 90°
- 45 Degree Elbow – where change in direction required is 45°

A 90 degree elbow is also called a "90 bend" or "90 ell". It is a fitting which is bent in such a way to produce 90 degree change in the direction of flow in the pipe. It used to change the direction in piping and is also sometimes called a "quarter bend". A 90 degree elbow attaches

readily to plastic, copper, cast iron, steel and lead. It can also attach to rubber with stainless steel clamps. It is available in many materials like silicone, rubber compounds, galvanized steel, etc. The main application of an elbow (90 degree) is to connect hoses to valves, water pressure pumps, and deck drains. These elbows can be made from tough nylon material or NPT thread.^[2]

A 45 degree elbow is also called a "45 bend" or "45 ell". It is commonly used in water supply facilities, food industrial pipeline networks, chemical industrial pipeline networks, electronic industrial pipeline networks, air conditioning facility pipeline, agriculture and garden production transporting system, pipeline network for solar energy facility, etc.^[2]

Most elbows are available in short radius or long radius variants. The short radius elbows have a center-to-end distance equal to the [Nominal Pipe Size](#) (NPS) in inches, while the long radius is 1.5 times the NPS in inches. Short elbows are widely available, and are typically used in pressurized systems.

Long elbows are typically used in low-pressure gravity-fed systems and other applications where low turbulence and minimum deposition of entrained solids are of concern. They are readily available in [acrylonitrile butadiene styrene](#) (ABS plastic), [polyvinyl chloride](#) (PVC) for DWV, sewage and central vacuums, [chlorinated polyvinyl chloride](#) (CPVC) and copper for 1950s to 1960s houses with copper drains

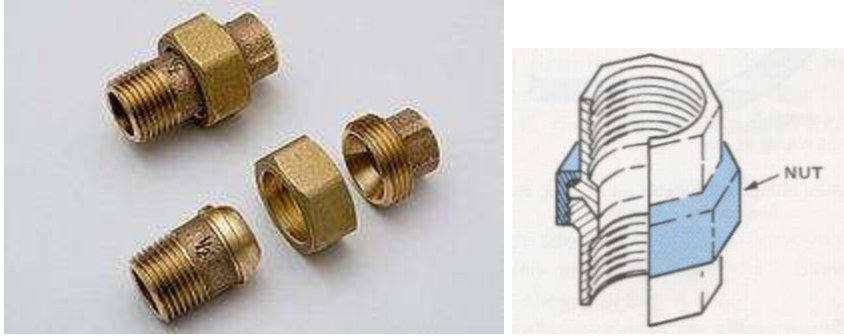
- **Coupling**



Pipe coupling (copper sweat)

A [coupling](#) connects two pipes to each other. If the size of the pipe is not the same, the fitting may be called a **reducing coupling** or **reducer**, or an **adapter**. By convention, the term "expander" is not generally used for a coupler that increases pipe size; instead the term "reducer" is used.

- **Union**



A combination pipe union and reducer fitting (brass threaded)

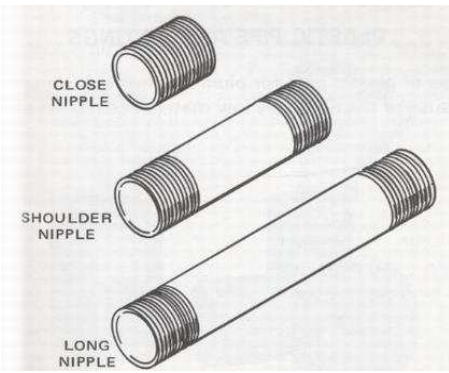
A union is similar to a coupling, except it is designed to allow quick and convenient disconnection of pipes for maintenance or fixture replacement. While a coupling would require either [solvent welding](#), [soldering](#) or being able to rotate with all the pipes adjacent as with a [threaded coupling](#), a union provides a simple transition, allowing easy connection or disconnection at any future time. A standard union pipe is made in three parts consisting of a nut, a female end, and a male end.^[2] When the female and male ends are joined, the nuts then provide the necessary pressure to seal the joint. Since the mating ends of the union are interchangeable, changing of a valve or other device can be achieved with a minimum loss of time. Pipe unions are essentially a type of flange connector, as discussed further below.

In addition to a standard union, there exist [dielectric](#) unions which are used to separate dissimilar metals (such as copper and galvanized steel) to avoid the damaging effects of [galvanic corrosion](#). When two dissimilar metals are in contact with an electrically conductive solution (even tap water is conductive), they will form a battery and generate a voltage by [electrolysis](#). When the two metals are in direct contact with each other, the [electric current](#) from one metal to the other will cause a movement of ions from one to the other, dissolving one metal and depositing it on the other. A dielectric union breaks the electric current path with a plastic liner between two halves of the union, thus limiting galvanic corrosion.

NIPPLES

Nipples are pieces of pipe, 12 inches (") or less in length threaded on both ends. They are used to join two fittings that are close together. These should be purchased because it is difficult to thread short pieces of pipe with conventional plumbing tools. Do not attempt to make them.

Nipples are specified by diameter and length. A *close nipple* is threaded along its entire length. *Shoulder nipples* have a short portion of unthreaded pipe. Length of a pipe nipple is determined by measuring from end to end.



Nipples are short connectors between

- **Reducer**



Reducer fittings, bronze threaded (left) and copper sweat (right)

A reducer allows for a change in pipe size to meet [hydraulic](#) flow requirements of the system, or to adapt to existing piping of a different size. Reducers are usually [concentric](#) but [eccentric reducers](#) are used when required to maintain the same top- or bottom-of-pipe level.



A **bushing** takes up the difference in diameter when a smaller pipe must be connected to a larger fitting.

Tee



Pipe tee (copper sweat)

A tee is the most common pipe fitting. It is available with all female thread sockets, all solvent weld sockets, or with opposed solvent weld sockets and a side outlet with female threads. It is used to either combine or split a fluid flow. It is a type of pipe fitting which is T-shaped having two outlets, at 90° to the connection to the main line. It is a short piece of pipe with a lateral outlet. A tee is used for connecting pipes of different diameters or for changing the direction of pipe runs. They are made of various materials and available in various sizes and finishes. They are extensively used in pipeline networks to transport two-phase fluid mixtures. They are categorized as:

- Equal
- Unequal

When the size of the branch is same as header pipes, equal tee is used and when the branch size is less than that of header size, reduced tee will be used. Most common are tees with the same inlet and outlet sizes. Some of the industrial tees are Straight Tee, Reducing Tee, Double Branch Tee, Double Branch Reducing Tee, Conical Tee, Double Branch Conical Tee, Bullhead Tee, Conical Reducing Tee, Double Branch Conical Reducing Tee, Tangential Tee, and Double Branch Tangential Tee.

The above tees are categorised on the basis of their shapes and structure. They can also be classified on the basis of the application they are required to perform.

- **Cross**

Cross fittings are also called 4-way fittings. If a branch line passes completely through a tee, the fitting becomes a cross. A cross has one inlet and three outlets, or vice versa. They often have

solvent welded socket ends or female threaded ends. Cross fittings can generate a huge amount of stress on pipe as temperature changes, because they are at the center of four connection points.



- **Cap**



Pipe cap (copper sweat)

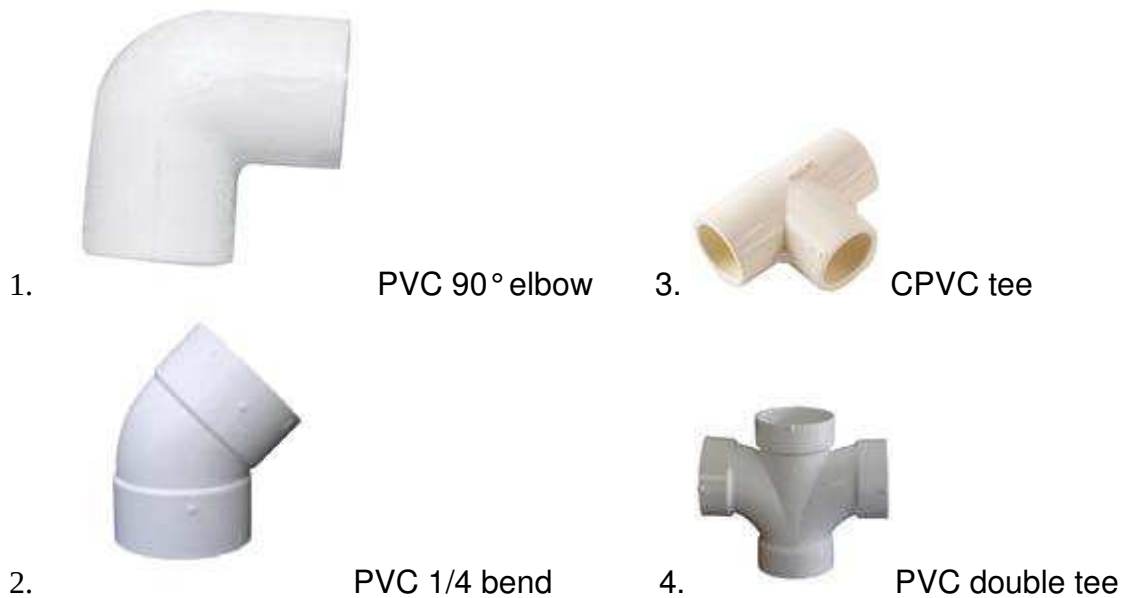
A type of pipe fitting, usually liquid or gas tight, which covers the end of a pipe. A cap is used like plug, except that the pipe cap screws or attaches on the male thread of a pipe or a nipple. A cap may have a solvent weld socket end or a female threaded end and the other end closed off. If a solvent weld cap is used to provide for a future connection point, several inches of pipe must be left before the cap. This is because when the cap is cut off for the future connection, enough pipe must remain to allow a new fitting to be glued onto it. In plumbing systems that use threads, the cap has female threads. Industrial caps can be round, square, rectangular, U-shaped, I-shaped and have a round hand grip or a flat hand grip.

- **Plug**

A plug closes off the end of a pipe. It is similar to a cap but it fits inside the fitting it is mated to. In a threaded iron pipe plumbing system, plugs have male threads. Some of the popular types of plugs are:

- Mechanical pipe plug
- Pneumatic disk pipe plug
- Single size pneumatic all rubber pipe plug
- Multi-size pneumatic pipe plug
- Multi-size flow-through pipe plug
- High pressure pipe plug

4.2 Different forms of pipe fittings used in pipe connections.





5.

copper double elbow



6.

copper wye



7.

copper double wye



8.

copper union



9.

copper female adapter



13.

copper male adapter



10.

galvanized steel 90°elbow



11.

chrome p-trap



12. black iron floor flange



14. black iron reducing tee



15. black iron 90° street elbow



16. galvanized steel 45° elbow



17. ABS closet flange 19. ABS wye



CHAPTER 4: PREPARING PIPING MATERIALS

1 CUTTING PIPES

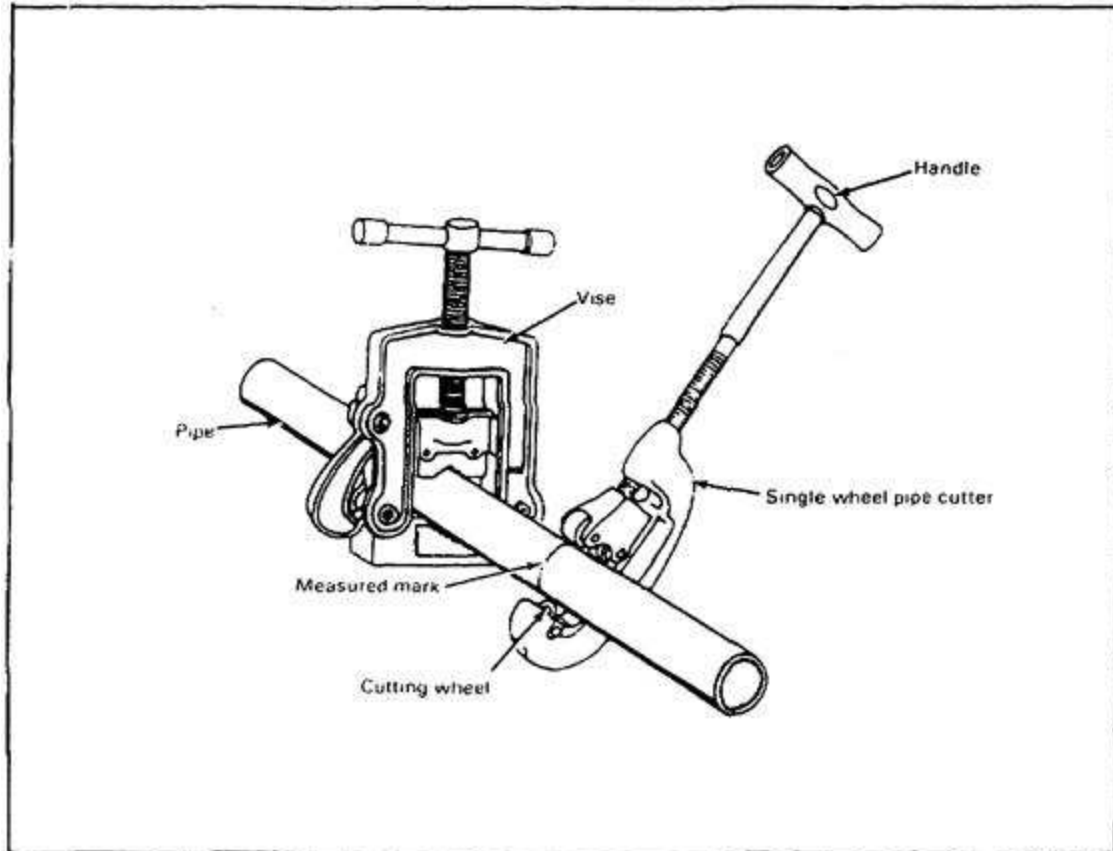
- **Tools for Cutting Metal Pipe**

- ❖ **Cutter pipe**

- **procedure:**

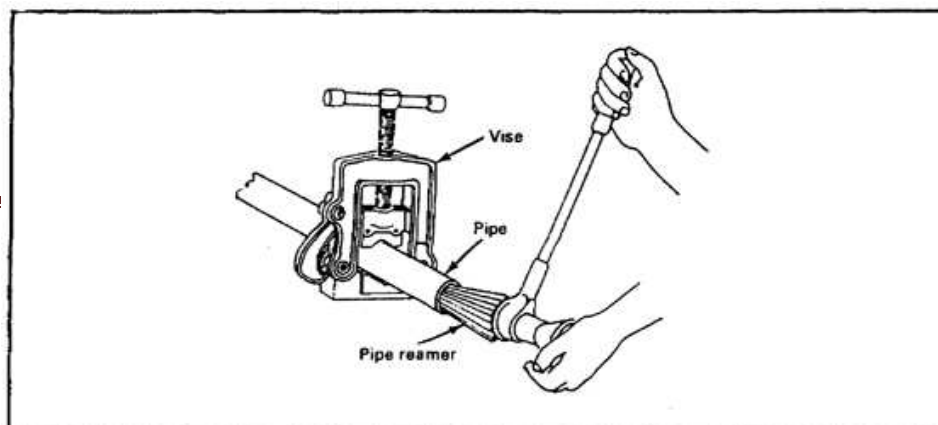
Place the pipe in a vise with the measured mark about eight inches from the vise. Set the cutting wheel on the mark and turn the cutter handle to get a bite on the pipe. To cut the pipe, turn the cutter clock wise and apply the cutting oil to the cutting wheel. After each complete turn, give the handle a quarter turn until the pipe is cut.

When pipe cutter is not available, use a hacksaw and insure the cut is as square as possible to simplify threading the pipe.



After cutting you have to remove burrs as follow as:

Removing Burrs. Push the reamer into the pipe and apply pressure with one hand. With the other hand, turn the reamer clockwise using short even strokes until all the burrs are removed from the cut end. Use a fine metal file to remove burrs from the outside of the pipe.



Hacksaw

Although many power and specialized tools exist for pipe cutting applications, the hack saw can be used to cut through soft metals and thin-walled pipes. The hack saw's three-sided metal frame connects to a pistol-grip handle. The frame looks like an upside-down "U" and a sharpened, multi-toothed blade attaches to its open side.



The hacksaw's changeable blade makes it convenient to cut through a variety of metal pipes.

Take a hacksaw to a thick-walled cast iron drain pipe and you're in for an excruciating experience; take a chain cutter to the same pipe and you can create a clean cut, no problem. The right cutting tools allow plumbers and pipe fitters to quickly and easily cut through both soft metal pipe, such as copper, and hard metal pipe, such as steel and iron. Whether you're cutting a water-supply pipe or drain line, familiarity with the types of metal pipe cutting tools allows you to choose the right tool for the job.

❖ Chain Cutter

The chain cutter cuts cast iron pipe, and is frequently used during drain line construction. The chain cutter tool scores and snaps cast iron pipe by means of a flexible chain attached to hinged

handles. The chain wraps around a pipe and is tightened by ratcheting from the handles. Each link contains a small cutting disc that eventually scores and snaps the pipe.

❖ **Tubing Cutter**

The tubing cutter is a hand tool that cuts soft metal pipe, such as copper, by running a sharpened carbide disc around the pipe's exterior. This tool features an adjustable mouth that opens to surround a piece of pipe and closes to press the cutting disc against the pipe's surface. Once tightened, you turn the cutter around the pipe's exterior until the disc shears through the pipe's wall.

❖ **Reciprocal Saw**

This power tool uses an electric motor to rapidly push sharpened blades back and forth in a sawing motion. Outfitted with metal cutting blades, the reciprocal saw cuts through all types of metal pipe, including cast-iron, copper and steel. The reciprocal saw performs rough cutting of installed pipe for demolition or repair, rather than cutting pipe to size for installation.

❖ **Angle Grinder**

The angle grinder's electric motor spins abrasive, circular blades to cut through metal pipe. The angle grinder's blade spins on an axis parallel to the tool's body, allowing you to cut with a forward pushing or side-to-side motion. The angle grinder, like the reciprocal saw, is usually used to remove pipe for repair or demolition; its unique design allows it to reach difficult spots and cut in awkward directions.

❖ **Chop Saw**

The chop saw, sometimes called "cut-off saw," cleanly cuts lengths of pipe to size. The chop saw's electric motor spins an abrasive, circular cutting disc that ranges from 8 to 12 inches or more in diameter. A retractable arm suspends the cutting blade above the saw's stationary base and lowers the blade through pipe placed on the base.

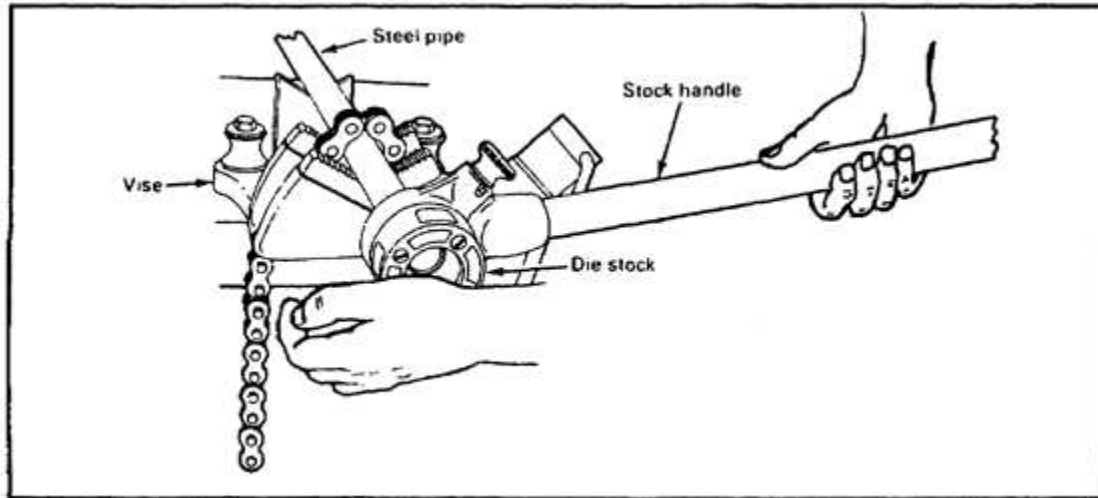
2 THREADING PIPES

- **PROCEDURE:**

Threading pipe ends.

Insert the correct size die into the die stock. Slide the die stock over the pipe and apply pressure with one hand. With the other hand, turn the stock handle slowly clockwise until the die has

taken a bite on the pipe. Apply cutting oil to the die as the stock handle is given one complete clockwise turn and backed off a quarter turn. Repeat this action until 1/4 inch of the pipe is beyond the die stock. The pipe is now threaded properly. Turn the handle in a counterclockwise direction to remove the pipe.





RATCHET PIPE THEADING

3 CONNECTIONS TYPES

- **Introduction**

Much of the work of installing a piping or plumbing system involves making leakproof, reliable connections. Depending on the technology used, basic skills may be required, or specialized skills and professional licensure may be required.

- ❖ **Fastener**

A fastener is a hardware device that mechanically joins or affixes two or more objects together. The following are the type of fasteners commonly used:

- Stud bolt with nut
- Machine bolt with nut

Usually the stud bolts are used with full threading and with two heavy hexagonal nuts.

❖ Threaded pipe



A **threaded pipe** is a pipe with a **screw thread** at one or both ends for assembly. Steel pipe is often joined using threaded connections, where tapered threads are cut into the end of the tubing segment. Threaded steel pipe is still widely used in many homes and businesses to convey **natural gas** or **propane** fuel, and is a popular choice in **fire sprinkler** systems due to its high heat resistance.

Assembling threaded steel pipe takes some skill, plus careful planning to allow lengths of pipe to be screwed together in proper sequence. Most threaded pipe systems require occasional use of pipe union fittings to allow final assembly.

❖ Solvent welding

A **solvent** is applied to PVC, CPVC, ABS, or other plastic piping, to partially dissolve and fuse the adjacent surfaces of piping and fitting. Solvent welding is usually used with a sleeve-type joint, to connect pipe and fittings made of the same (or closely compatible) material.

Unlike regular welding of metals, solvent welding is relatively easy to perform, although care is still needed to produce reliable joints. Solvents typically used for plastics are usually **toxic**, may be **carcinogenic**, and may also be **flammable**, requiring adequate ventilation.

❖ Soldering

To make a **solder** connection, a chemical **flux** is applied to the inner sleeve of a sleeve type joint, and the pipe is inserted. The joint is then heated using a **propane torch**, solder is applied to the heated joint, and the melted solder is drawn into the joint by **capillary action** as the flux vaporizes. **Sweating** is an alternate term sometimes used to describe soldering of pipe joints.

In situations where many connections must be made in a short period of time (such as plumbing of a new building), solder offers much quicker and much less expensive joinery than compression or flare fittings. A degree of skill is needed to rapidly make large numbers of reliable soldered joints. If flux residues are thoroughly cleaned, soldering can produce a long-lasting connection at low cost. However, the use of open flames for heating joints can present

fire and health hazards to occupants of a building being worked on, and requires adequate ventilation.

❖ **Brazing**

Brazing is a thermal joining process in which the two pieces of the base metal are joined when a molten brazing filler metal is allowed to be drawn into a capillary gap between them.

It is used to join most metals and alloys commonly used in engineering. Brazing filler metals have very high melting points, but always below the melting point of the metals being joined. It is closely related to soldering, but uses harder materials and higher temperatures. Brazing can be used to join pipes, rods, metal pieces or any other shape as long as the pieces fit neatly against each other without leaving large gaps. It is also capable of joining **tungsten carbide**, ceramics and similar non-metallic materials.

Successfully brazed joints are as strong as the parent metal pieces being joined and can withstand demanding service conditions i.e they are strong and ductile. Well brazed joints have smooth, neat fillets and they offer good electrical conductivity.

❖ **Welding**

Welding of metals differs from soldering and brazing, in that the connection is made without adding a special low-melting-point material (e.g. solder) to complete a joint. Instead, the material of the pipe or tubing is itself partially melted in a carefully controlled manner, and the fitting and piping are directly fused together. Generally this requires that the piping and the fitting be made of the same (or closely compatible) material. Pipe welding is a specialized skill, and is often performed by specially licensed workers who are tested periodically for the quality of their work. For ultra-critical applications, every joint is tested using **non-destructive testing** methods.

Properly welded and inspected joints are considered to be very reliable, robust, and long-lasting. Proper ventilation is essential to remove dangerous metallic fumes from welding operations. Because of the expensive skilled labor required, welded pipe joints are usually restricted to high-performance applications, such as in **shipbuilding**, **chemical reactors**, and **nuclear reactors**.

4.PIPES

You may classify pipes according either to their uses or materials in which are manufactured for or their sizes (nominal diameters)

Cast iron pipes: these types of pipes are used only for low pressure and are well suited for water, gas and sewage

Wrought iron (malleable iron pipe): are used for carrying water and gas at low pressure for domestic purposes (are galvanized by dipping them in molten zinc).

Steel pipes(there are in three categories ;low carbon steel, medium and high carbon steel):these pipes are mainly used for carrying water ,steam, gas ,sewage and air at a high pressure; compare to others above.

Copper and Brass pipes: are mainly used for hot water systems, refrigeration system and general plumbing work

Aliminium pipes: these types are used in aircraft and marine applications because of its light weight.

Lead pipes: are generally used for sanitary purposes, the joints are made by soldering process.

Concrete pipes: are commonly used for water mains, sewers and in irrigation.

*Plastic pipes (PVC):*these are cheap and light in weight ,are extensively used for sanitary purpose and electric wire conduct

Rubber pipes: are flexible and available in coils, are used in household work for water, oil and gas.

4 STANDARD PIPE SIZES COMMON USED IN PIPING

The table below shows nominal diameter of pipes

Imperial system (inches)	Metric system (in mm)	Imperial system (inches)	Metric system (in mm)
$\frac{1}{4}$	8	3	76
$\frac{1}{2}$	13~15	$3 \frac{1}{4}$	82
$\frac{3}{4}$	20	$3 \frac{1}{2}$	90
1	25	4	100
$1 \frac{1}{4}$	32	$4 \frac{1}{4}$	108
$1 \frac{1}{8}$	28	$4 \frac{3}{8}$	110
$1 \frac{1}{2}$	40		
$2 \frac{1}{4}$	60		

